Rethinking Shape: Developing a Novel Circular Object Detection Method using Deep Learning

Master Thesis

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Abstract

THIS IS MY ABSTRACT

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Chapter 1

Introduction

1.1 Background information on object detection in agriculture

In modern agriculture, accurate detection and size estimation of fruits and vegetables play an important role in automating crop monitoring, yield prediction, and selective harvesting. Existing deep learning approaches for object detection, such as YOLO and RF-DETR , rely on rectangular bounding boxes to localize objects in images. While effective for general-purpose detection tasks, these rectangular representations fail to capture the shape of many natural circular-shaped objects, including fruits and vegetables. A practical challenge when applying rectangular boxes on circular-shaped objects is that the output shape will contain redundant background pixels and that it doesn't mimic its real shape leading to inaccurate localization and size estimation, especially in the presence of occlusion.

1.2 Research questions

This thesis proposes a novel approach for circular object detection by rethinking the geometric representation used in deep learning architectures. The central research question guiding this thesis is: *Can the prediction of naturally circular objects using a circle-shaped object detection model be improved compared to a standard rectangular-shaped object detection model?*

To address this question, the project aims to design and evaluate a neural network that predicts circular parameters, which are the centre coordinates and the radius of objects, instead of the traditional bounding boxes. By integrating a circular representation into the prediction head of a detection model, the model is expected to achieve more precise detection and more reliable size estimates, as well as decrease the computational power needed as the number of parameters is lowered.

1.3 General methodology

In more detail, to answer the research will involve (1) constructing a circular detection head compatible for the YOLOv1 object detection model, (2) defining a suitable loss function for circle regression, and (3) benchmarking the performance against the conventional box-based-models using the broccoli's dataset of Blok, van Henten, van Evert and Kootstra.

1.4 Scope of research

1.4.1 Model choice

This research focuses specifically on modifying the YOLOv1 architecture rather than a more recent state-of-the-art model such as RF-DETR. Although RF-DETR has demonstrated superior performance on broccoli detection tasks, there are three motivating reasons for this choice.

First, the relative simplicity of YOLOv1's architecture makes the integration of a circular prediction head considerably more manageable. RF-DETR's transformer-based design introduces additional complexity in the prediction pipeline that would make isolating and replacing the bounding box representation substantially more difficult.

Second, the field of object detection evolves at a rapid pace, with new architectures being released on a near-continuous basis. Anchoring this research to the current state-of-the-art models carries the practical risk that the chosen model becomes outdated within the project's timeframe. YOLOv1, by contrast, is a well-established and stable baseline whose detection capabilities are thoroughly documented and understood.

Third, and most importantly for the validity of this study, YOLOv1 is a comparatively transparent model. More recent architectures such as RF-DETR incorporate attention mechanisms, large pre-trained backbones, and complex post-processing steps that introduce confounding factors when attempting to isolate the effect of a single architectural change. By working with a simpler model, the performance difference between circular and rectangular representations can be attributed more directly to the geometric change itself, rather than to incidental interactions with other components of the architecture.

Chapter 2

Related Works

2.1 Image-based size estimation of broccoli heads under varying degrees of occlusion (Blok, van Henten, van Evert, & Kootstra, 2021 [2])

This paper discusses how they dealt with estimating broccoli heads under partial occlusion. As can be seen in Figure 2.1, this was done by first predicting a bounding box using the ORCNN algorithm. After that, they predicted 2 masks within the bounding box, one for the visible region of the broccoli and one for the amodal region (amodal is the combined visible and non-visible region belonging to an object of interest). This combined output of the amodal region demonstrated an improvement of the size estimate of the broccoli heads as an input for selective harvesting. Granted this is a positive result, the complex neural network architecture proved to need a lot of input data to optimize efficiently.

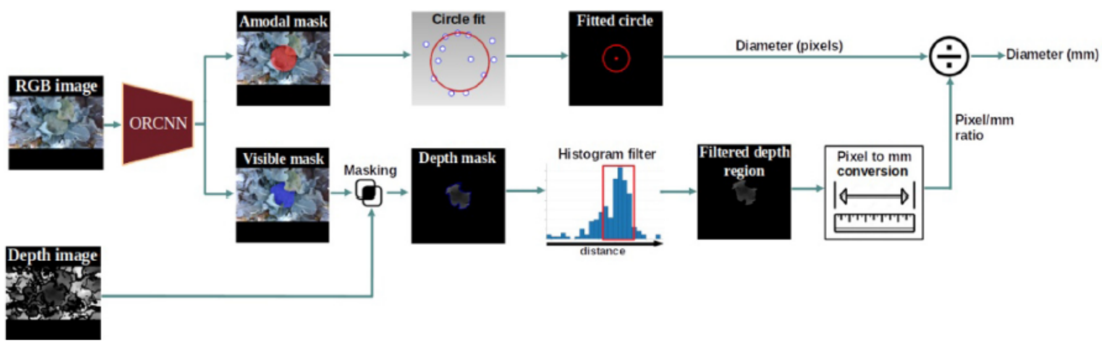


Figure 2.1: Schematic representation of the diameter estimation method using the segmentation output of ORCNN

2.2 BubCNN: Bubble detection using Faster RCNN and shape regression network (Haas, Schubert, Eickhoff, & Pfeifer, 2020) [4]

This study has the same idea. As can be seen in Figure 2.2, a two-stage deep learning system is used for detecting and measuring gas bubbles in multiphase flows. It first locates bubbles in images by predicting bounding boxes around the bubbles using a faster R-CNN detector. After that, a CNN for shape regression approximates the shape of each bubble as an ellipse (using the center, major/minor axes, and rotation). This hybrid design integrates the localization efficiency of region-based detectors with the geometric precision of a regression network, allowing automatic generalizable bubble analysis under diverse experimental conditions.

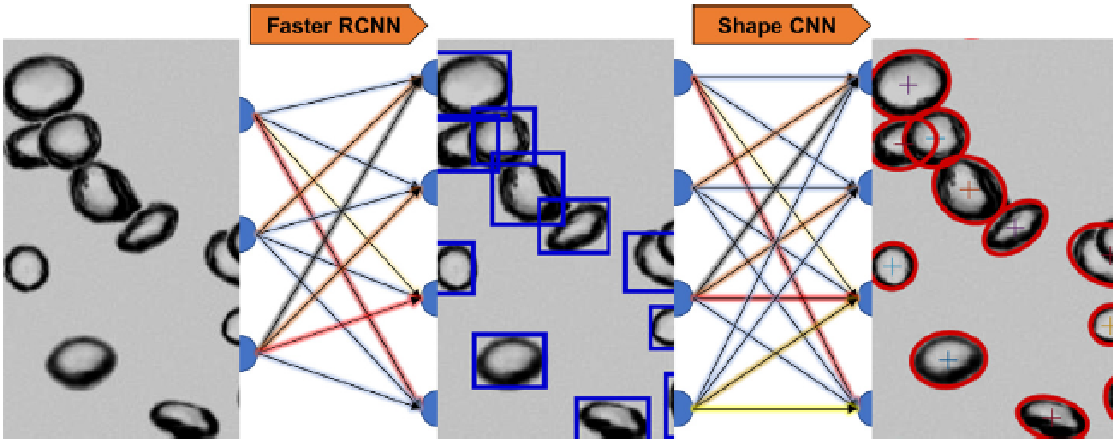


Figure 2.2: Schematic representation of the BubCNN architecture

2.3 Circle representation for Medical Object Detection (Nguyen, et al., 2021) [7]

As for Circle representation for Medical Object Detection (Nguyen, et al., 2021), CircleNet is an anchor-free deep learning model designed to detect ball-shaped biomedical objects (like glomeruli or nuclei) more efficiently than traditional box-based detectors. Instead of predicting rectangular bounding boxes (x, y, width, height), CircleNet predicts circular regions described by the center point (cx, cy) and the radius r. This design makes CircleNet rotation-invariant, simpler (fewer degrees of freedom), and better aligned with naturally round biological objects. In essence, this paper discusses almost the technology as we want to experiment with, except it builds on the framework of CenterNet and is focused on the biomedical sector, whereas this thesis applies a comparable circular representation to the YOLOv1 architecture. While CircleNet demonstrates that a circular output head can be integrated into a modern anchor-free detector, YOLOv1's comparatively transparent architecture makes it easier to reason about the contribution of the geometric representation change specifically. Additionally, extending the circle representation to a simpler and more widely understood baseline broadens the evidence that circular detection is a generally applicable principle, rather than one tied to the particular design choices of CenterNet.

Chapter 3

Methods

This chapter describes the methodology used to develop and evaluate a circular object detection model based on the YOLOv1 architecture. Section 3.1 describes the dataset. Section 3.2 introduces the baseline YOLOv1 model. Sections 3.3 through 3.9 then detail each modification made to support circular predictions, covering the prediction head, target encoding, loss function, numerical stability, post-processing, and evaluation protocol.

3.1 Dataset

All experiments use the publicly available broccoli dataset introduced by Blok, van Henten, van Evert, and Kootstra [2]. The dataset consists of RGB images of broccoli heads captured in field conditions, annotated with circle labels in LabelMe format. Each annotation specifies a centre point (c_x, c_y) and a second point on the perimeter, from which the radius r is derived as the Euclidean distance between the two points. The dataset is divided into training, validation, and test splits, with each image stored as a `.png` file paired with a `.json` annotation file.

Images are resized to 448×448 pixels prior to all experiments. During training, the following data augmentations are applied: random horizontal flipping with probability 0.5; random affine transformations with scale sampled from $[0.8, 1.2]$ and translation up to $\pm 20\%$ of the image dimensions; and random colour jitter affecting brightness, contrast, saturation, and hue. Circle annotations are processed through spatial augmentations by tracking the circle centre as a keypoint and the enclosing bounding square as a bounding box. After transformation, the radius is recomputed as the mean of half the transformed bounding box width and half the transformed bounding box height, which approximates the true radius under the applied distortions. No augmentation is applied during validation or testing.

3.2 Baseline YOLOv1 Architecture

The baseline model follows the YOLOv1 detection model [8], adapted for a single-class setting. The architecture consists of three components.

The **backbone** is a ResNet-34 network [5] pretrained on ImageNet, which extracts a 14×14 feature map with 512 channels from a 448×448 input image.

The **detection head** consists of four sequential blocks of a convolutional layer, batch normalisation, and Leaky ReLU activation with slope 0.1. A strided convolution in the second block reduces the spatial resolution from 14×14 to 7×7, yielding a $7 \times 7 \times 1024$ feature map.

The **prediction layer** is a 1×1 convolution that maps the 1024-channel feature map to a prediction tensor of shape $S \times S \times (P \cdot B + C)$, where grid size $S = 7$, the number of predictors per cell is $B = 2$, the number of classes is $C = 1$, and the number of parameters per predictor is P. In the baseline, each predictor encodes a bounding box as $(x_{\text{center}}, y_{\text{center}}, \sqrt{w}, \sqrt{h}, \text{conf})$, so $P = 5$.

3.3 Circular Prediction Head

The central modification of this thesis is replacing the rectangular parameterisation with a circular one. Rather than predicting four geometric parameters per box, each predictor outputs three geometric parameters and a confidence score:

$$(x_{\text{center}},\ y_{\text{center}},\ \sqrt{r},\ \text{conf})$$

giving $P = 4$. The prediction tensor therefore has shape $S \times S \times (4 \cdot B + C)$, reducing the output dimensionality by B values per cell compared to the rectangular baseline.

The model predicts, for each grid cell at column i and row j and predictor b:

- x_{center}: the x-offset of the circle centre relative to the cell's left edge, as a fraction of cell width,
- y_{center}: the y-offset of the circle centre relative to the cell's top edge, as a fraction of cell height,
- $\sqrt{r}$: the square root of the radius normalised to the image size, following the same transformation used for box dimensions in the original YOLO formulation to prevent large radii from dominating the loss,
- conf: the confidence score.

The absolute centre coordinates are recovered as:

$$c_x = \frac{x_{\text{center}}}{S} + \frac{i}{S}, \qquad c_y = \frac{y_{\text{center}}}{S} + \frac{j}{S} \tag{3.1}$$

and the radius as $r = (\sqrt{r})^2$. These coordinates are normalised to $[0, 1]$ relative to the image dimensions.

3.4 Circle Target Encoding

To construct ground-truth targets in the YOLO grid format, each annotated circle (c_x, c_y, r) is assigned to the grid cell (i, j) containing the circle centre. The image of height H and width W is divided in $S \times S$ grid, so each cell spans H/S pixels vertically and W/S pixels horizontally. The grid cell coordinates are then:

$$i = \left\lfloor \frac{c_x^{\text{px}}}{W/S} \right\rfloor, \qquad j = \left\lfloor \frac{c_y^{\text{px}}}{H/S} \right\rfloor \tag{3.2}$$

where c_x^{px} and c_y^{px} are the circle centre coordinates in pixels. As we know from 3.1, $H = W = 448$ and $S = 7$, so each cell covers 64×64 pixels. Within such a grid cell, the target vector for all B predictors is set to:

$$\left(x_{\text{center}},\ y_{\text{center}},\ \sqrt{r/\max(W, H)},\ 1.0\right) \tag{3.3}$$

where x_{center} and y_{center} are the intra-cell offsets of the centre, W and H are the image width and height after resizing, and the confidence target is 1.0 for all cells containing an object. For all remaining cells, the target vector is the zero vector.

3.5 Circle Intersection over Union

Selecting the responsible predictor and computing the objectness regression target during training requires a differentiable measure of overlap between two circles. Given two circles with centres (c_{x1}, c_{y1}) and (c_{x2}, c_{y2}) and radii r_1 and r_2, let $d = \sqrt{(c_{x1} - c_{x2})^2 + (c_{y1} - c_{y2})^2}$ be the distance between their centres. Three geometric cases are handled analytically:

1. **No overlap** ($d \geq r_1 + r_2$): the intersection area is zero.
2. **Containment** ($d \leq |r_1 - r_2|$): one circle lies entirely within the other; the intersection equals the area of the smaller circle, $\pi \min(r_1, r_2)^2$.
3. **Partial overlap** (otherwise): the intersection is computed via the circular segment formula:

$$I = r_1^2 \arccos\left(\frac{d^2 + r_1^2 - r_2^2}{2\,d\,r_1}\right) + r_2^2 \arccos\left(\frac{d^2 + r_2^2 - r_1^2}{2\,d\,r_2}\right) \\ - \tfrac{1}{2}\sqrt{(-d + r_1 + r_2)(d + r_1 - r_2)(d - r_1 + r_2)(d + r_1 + r_2)} \tag{3.4}$$

The circle IoU is then $\mathrm{IoU}_\circ = I \,/\, (A_1 + A_2 - I)$, where $A_k = \pi r_k^2$. This follows the derivation of Nguyen et al. [7] in a different arccos form. In formula 3.4, cosine arguments are clamped to $[-1, 1]$ and the term under the square root is clamped to zero from below to ensure numerical stability.

3.6 Loss Function

The overall loss follows the multi-part YOLO formulation, with the localisation term adapted for circles. Given a batch of size N, the total loss is:

$$\mathcal{L} = \frac{1}{N}\Big[\lambda_{\text{coord}}\big(\mathcal{L}_x + \mathcal{L}_y + \mathcal{L}_r\big) + \mathcal{L}_{\text{obj}} + \lambda_{\text{noobj}}\,\mathcal{L}_{\text{noobj}} + \mathcal{L}_{\text{cls}}\Big] \tag{3.5}$$

where $\lambda_{\text{coord}} = 5$ and $\lambda_{\text{noobj}} = 0.5$, matching the original YOLO weighting. For each cell, the responsible predictor b^* is defined as the one whose predicted circle achieves the highest circle IoU with the ground-truth circle. To compute this IoU, the raw cell offsets are first converted to absolute normalised coordinates using equation 3.1. The localisation loss terms $\mathcal{L}_x$, $\mathcal{L}_y$ and $\mathcal{L}_r$ however operate directly on raw cell offsets, consistent with the original YOLO formulation. The individual loss terms are then:

Wait the sqrt(r) and (x, y) losses have a different dimension. Same for the conf and class thing unless you do some sort of normalisation.
Thisi is probably already in the YoLo thing but how is it normalised?

$$\mathcal{L}_x = \sum_{i,j} \mathbf{1}_{ij}^{\text{obj}}\,(x_{\text{center}}^{b^*} - \hat{x}_{\text{center}})^2 \tag{3.6}$$

$$\mathcal{L}_y = \sum_{i,j} \mathbf{1}_{ij}^{\text{obj}}\,(y_{\text{center}}^{b^*} - \hat{y}_{\text{center}})^2 \tag{3.7}$$

$$\mathcal{L}_r = \sum_{i,j} \mathbf{1}_{ij}^{\text{obj}}\,\big(\sqrt{r}^{\,b^*} - \sqrt{\hat{r}}\big)^2 \tag{3.8}$$

$$\mathcal{L}_{\text{obj}} = \sum_{i,j} \mathbf{1}_{ij}^{\text{obj}}\,\big(\text{conf}^{b^*} - \mathrm{IoU}_\circ^{b^*}\big)^2 \tag{3.9}$$

$$\mathcal{L}_{\text{noobj}} = \sum_{i,j}\sum_{b}\big(1 - \mathbf{1}_{ij}^{\text{obj}} \cdot \mathbf{1}_b^{b^*}\big)\big(\text{conf}^b\big)^2 \tag{3.10}$$

$$\mathcal{L}_{\text{cls}} = \sum_{i,j} \mathbf{1}_{ij}^{\text{obj}}\,(p^{b^*} - \hat{p})^2 \tag{3.11}$$

where $\mathbf{1}_{ij}^{\text{obj}}$ is an indicator that cell (i, j) contains an object, and $\hat{\cdot}$ denotes the ground-truth value. The objectness target for the responsible predictor is set to the circle IoU between the predicted and ground-truth circles, consistent with the original YOLO formulation.

3.7 Numerical Stability

As mentioned in subsection 3.5, during training the arguments of the arccos terms in the partial overlap case are derived from predicted values that may contain small floating-point errors. Three measures are taken to address this.

First, the circle IoU value used as the objectness regression target is detached from the computational graph before the loss is computed, preventing gradients from flowing through the arccos branch.

Second, gradient clipping with a maximum ℓ_2 norm of 1.0 is applied to all parameters before each optimiser step.

??? L_∞ norm?

Third, the training loop checks whether all parameter gradients are finite before each optimiser step. If non-finite gradients are detected, the optimiser step is skipped for that batch and the batch tensors are saved to disk for offline inspection, but training continues uninterrupted, preventing the model weights from corrupting.

3.8 Post-processing

At inference time, the model output is decoded from the $S \times S$ grid back to circle coordinates (c_x, c_y, r) in normalised image space. The confidence score thresholds were selected by running a threshold-tuning script that sweeps over the candidate values and checks the F1 score on the validation set. This yielded a visualisation threshold of 0.2, an mAP evaluation threshold of 0.001, and an NMS threshold of 0.6. It is important to note here that the threshold tuning was solely done on the baseline model and kept the same for the new circular model, to ensure that differences in evaluation performance between the two models are attributable solely to the change in geometric representation rather than threshold optimisation.

define your abbreviations

Non-Maximum Suppression is then applied using true circle IoU, computed as described in subsection 3.5. Predictions are sorted by decreasing confidence score, and each prediction is suppressed if its circle IoU with another already-retained and a higher confidence prediction meets or exceeds the threshold of 0.6. This procedure replaces the bounding square used in the baseline, ensuring that the circle model uses circle IoU consistently across training loss, NMS and evaluation matching.

3.9 Evaluation Protocol

Both models are evaluated on the test set using mean Average Precision (mAP) at an IoU threshold of 0.5, consistent with the PASCAL VOC evaluation protocol [3] and the original YOLOv1 evaluation [8]. The two models differ in how IoU is computed during evaluation, reflecting the geometry of their respective output representations.

The baseline bounding-box model is evaluated using standard rectangular IoU. The circular model is evaluated using circle IoU, computed, as described in subsection 3.5, between the predicted circles and the ground-truth circle annotations. The reasoning for applying different measures is that applying box IoU to circle predictions would systematically underestimate their overlap. A perfectly predicted circle would have box IoU of $\pi/4 \approx 0.785$ with its ground-truth bounding square instead of 1.0, which would unfairly penalise the circle model. This means both models are fully symmetric in their use of IoU: the baseline uses rectangular IoU for training, NMS and evaluations, while the circular model uses circle IoU for all three.

No the IoU measures are
e in both cases and they don't penalise
the circle, you just
e the area differently (as appropriate)
ays: we expect an improvement for circles
nd the best we can hope for is
IoU /rect iOu = 4/π. This remark is
ablly best in the intro somewhere.

For both models, average precision is computed by sorting all test-set predictions by decreasing confidence, matching each prediction to the highest-IoU unmatched ground-truth annotation for its class, and marking it as a true positive if the IoU exceeds 0.5. The area under the precision-recall curve is computed using the area method, in which the curve is first monotonically smoothed by replacing each precision value with the maximum precision at any equal or higher recall, after which the area is obtained by summing the rectangular areas between consecutive recall changes.

3.10 Training Configuration

Both models are trained with SGD using momentum 0.9 and weight decay 5×10^{-4}. Both use the same learning rate schedule: a linear warmup over the first 3 epochs, increasing from 10% to

This basically a Tuckey window. Make a little plot of it

100% of the peak learning rate $\eta_{\max} = 0.001$, followed by cosine annealing [6] according to:

$$\eta_t = \eta_{\min} + \frac{1}{2}(\eta_{\max} - \eta_{\min}) \left(1 + \cos\left(\frac{\pi\,(t - t_{\text{warmup}})}{T - t_{\text{warmup}}}\right)\right) \tag{3.12}$$

where t is the current epoch, $t_{\text{warmup}} = 3$, and $\eta_{\min} = 1 \times 10^{-5}$.

The baseline model is trained for a fixed $T = 30$ epochs. The circular model is trained for up to $T = 50$ epochs with early stopping: training halts if the validation loss does not improve by at least 0.001 over 8 consecutive epochs. For both models, the checkpoint with the lowest validation loss is used for all evaluations. Training and validation loss, together with validation mAP, are recorded after each epoch to a JSON log file. These logs are used to produce training curves for both models, which are used in Chapter 4. All experiments use a batch size of 4 and run on a single GPU.

Chapter 4

Results

In this chapter the experimental results for both the baseline bounding-box model and the proposed circular model will be presented. In section 4.1 we will report on the quantitative performance on the test set for both models. As for section 4.2, we will analyse the training dynamics for both models. Last but not least, in section 4.4, qualitative examples of model predictions on test images will be presented.

4.1 Quantitive performance

Both models were evaluated on the test set of 487 images as described in subsection 3.9. The results of these evaluations are summarized in table 4.1. As can be seen, the new circular model achieves a mAP of 0.7598, compared to the 0.7038 mAP of the baseline. This is an improvement of 0.056 (approximately +8.0%).

And this is the point where you compare with π/4 = 0.785 (this is the optimum the baseline can get, vs 1.0 that a circular broccoli can get.

Model	**IoU metric**	**Train loss**	**Val loss**	**Best epoch**	mAP
Baseline YOLOv1	Rectangular IoU	0.2450	0.2984	49	0.7038
Circular YOLOv1	Circle IoU	0.4823	0.5196	45	0.7598

Table 4.1: Model training and test data performance

4.2 Training dynamics

As can be seen in Figure 4.1, the loss curves (in the left figure) follow a similar trajectory for both models. A sharp drop can be seen in the first 5 epochs, where the first 3 epochs are driven by the warmup schedule and after that the learning rate is driven by cosine annealing. After the sharp drop, both models gradually converge. The baseline reaches a final train/validation loss of 0.2450/0.2984, while the circle model settles notably higher at 0.4823/0.5196. The two models' loss values are not directly comparable, as they are computed from fundamentally different loss functions with different mathematical formulations. What matters for the comparison is the mAP on the test data, reported in subsection 4.1.

Not 1 to 1 but they do seem quite similar. The percentage rate of loss is more comparable

Certainly matters more!

The validation mAP curves (right plot in Figure 4.1) shows that the baseline mAP initially climbs faster, getting to around 0.5 mAP at epoch 10 while the circle mAP lags behind. From around epoch 15, both models have around the same mAP, with the circle model getting ahead from around epoch 22. After epoch 42, both models seem to plateau and stabilize at a validation mAP of 0.71.

However, you do claim that the circle method works a little better!, even if they end up with the same mAP if you train long enough.

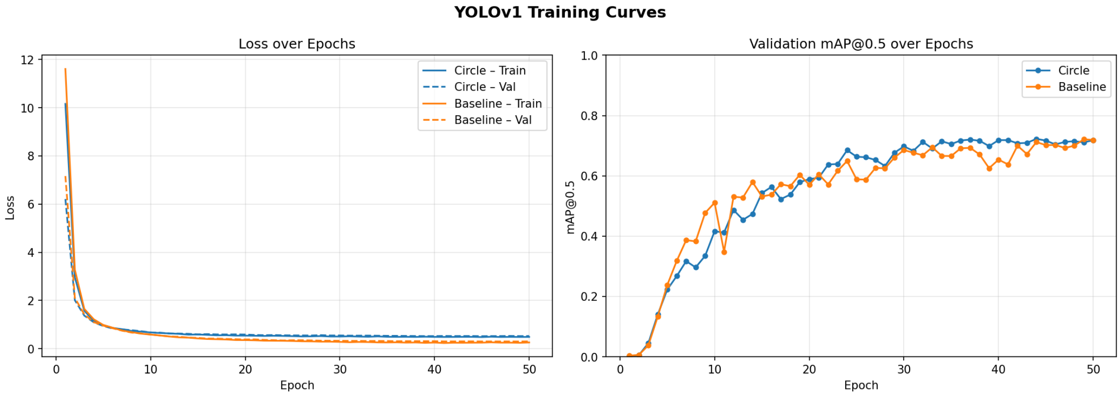


Figure 4.1: Training curves of both models

4.3 Precision-recall analysis

Figure 4.2 shows the precision-recall curves (PR-curves) for both models on the test data. The two PR-curves show us different detection behaviours. Firstly, the baseline model maintains a very high precision (≈ 1.0) up to a recall of around 0.6, after which the precision drops very fast, reaching almost zero at a recall of ≈ 0.73. This tells us that the baseline model is highly precise when it does find a broccoli head, but fails to find a substantial fraction of the ground-truth broccoli heads.

The circular shows a different PR-curve. The precision declines more gradually across the recall, remaining above 0.80 precision up to a recall of ≈ 0.70 and above 0.60 with a recall of ≈ 0.83. The maximum recall is approximately 0.90, compared to the 0.73 for the baseline. The circular model therefore retrieves a larger fraction of the ground-truth heads, trading the good early precision of the baseline for a extended recall range. The net effect is a larger area under the curve, with AP = 0.760 for the circular model compared to the baseline's 0.704 AP.

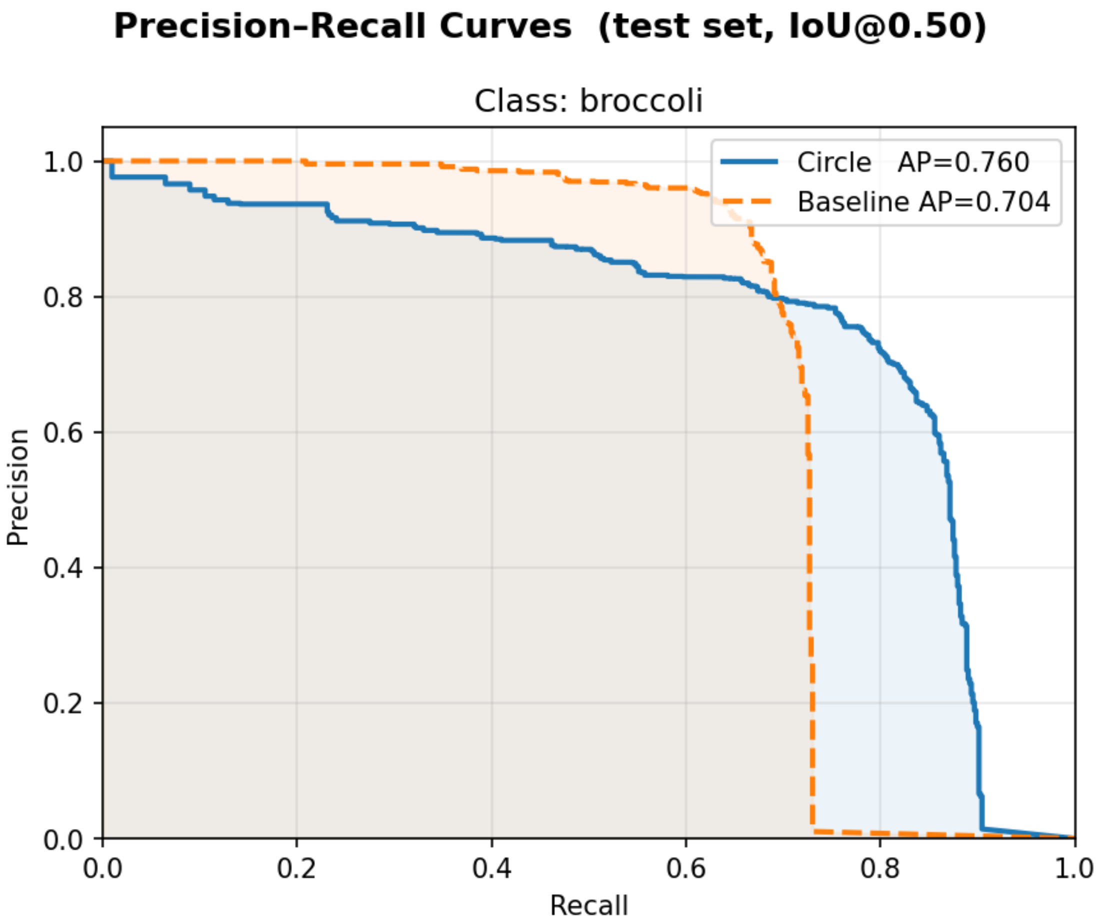


Figure 4.2: Precision-Recall curves

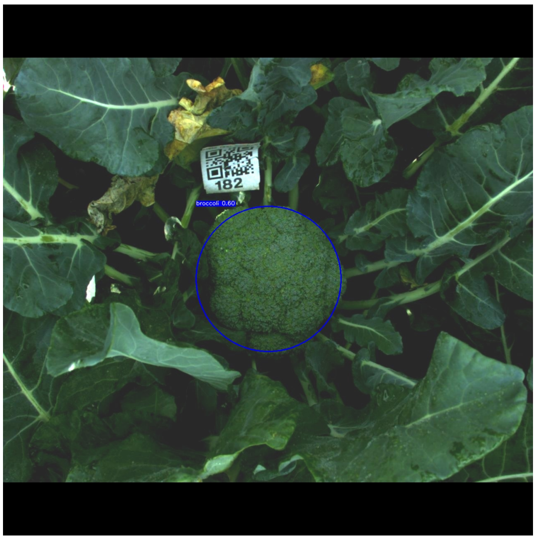
4.4 Qualitative results

In this section, the models will be qualitatively analysed by comparing the baseline model's predictions to the circle model's predictions on the same test set of 10 sample images.

4.4.1 Boundary fit

Isolated head On clearly visible, unoccluded broccoli heads, as can be seen in Figure 4.3, for the top 2 images, the circular model's predicted boundary follows the contour of the floret more closely. For the bottom 2 images, however, the baseline model's predicted boundary seems to sit more tightly around the broccoli head. Due to the geometric advantage of the circle representation on the broccoli head, the circles seem to capture less additional background information.

Partially occluded head As can be seen in Figure 4.4, the two models show a slightly different qualitative response. The baseline tends to fit its bounding box tightly around the visible part of the heads and overestimates the occluded part of the heads. It seems that the bounding box extends noticeably into the surrounding leaf area because the box edges must be placed somewhere and the occluded part of the head does not provide a signal. The circular model, in comparison, produces predictions that closely align with the visible circular region, as the circle boundary naturally conforms to the round contour of the floret. This is consistent with the observation by



Circular model (conf. 0.60)

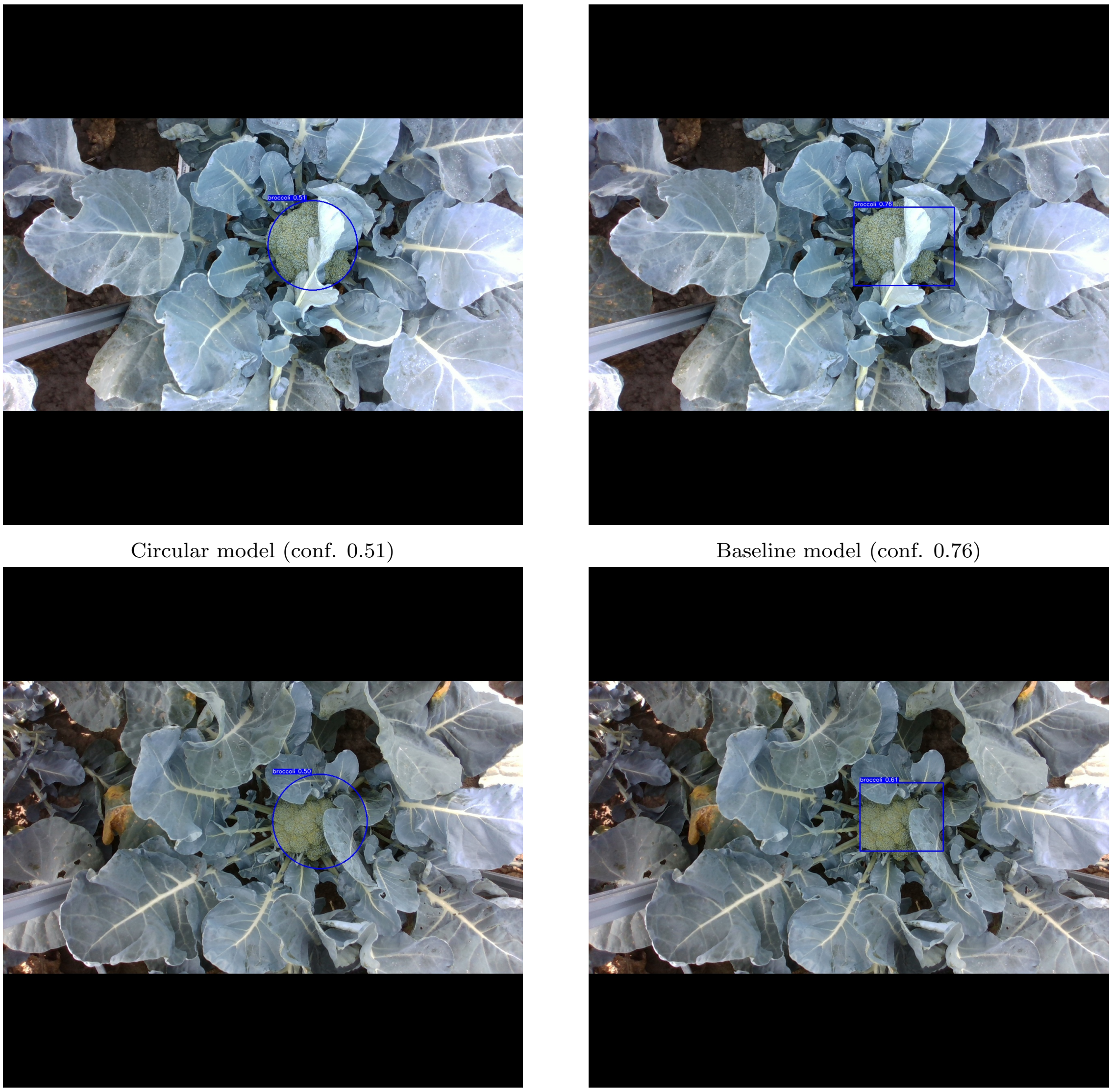
Baseline model (conf. 0.92)

Circular model (conf. 0.61)

Baseline model (conf. 0.82)

Figure 4.3: Isolated head - boundary fit comparison

Blok et al. [2] that broccoli heads are well-approximated by circular shapes, and thus the circular predictions align more closely to the ground truth.



Circular model (conf. 0.51) Baseline model (conf. 0.76)

Circular model (conf. 0.50) Baseline model (conf. 0.61)

Figure 4.4: Partially occluded head - boundary fit comparison

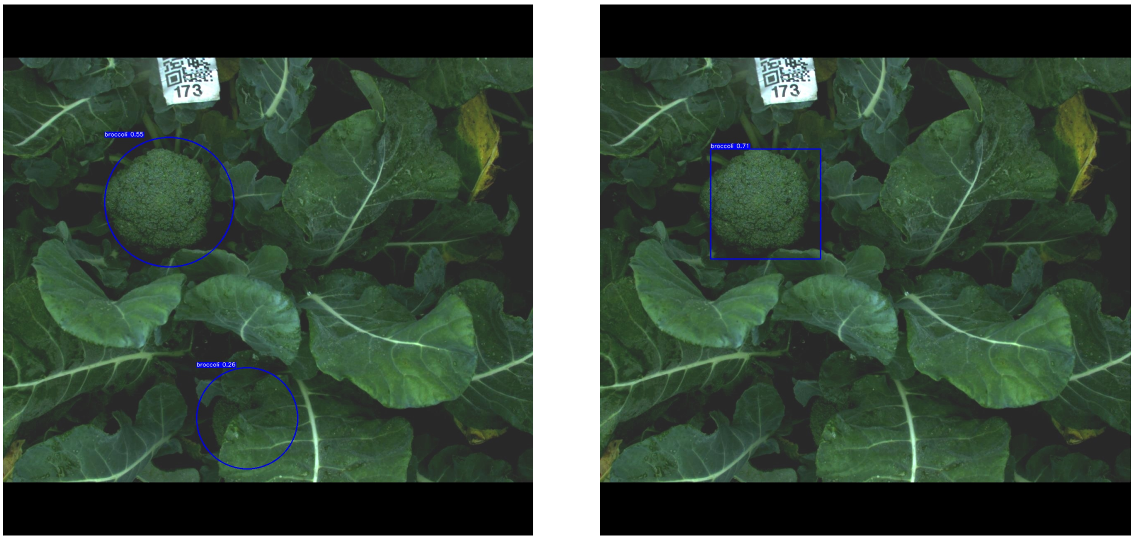
THe confidences are consistently a lot lower!!
Any ideas why?

4.4.2 Detection of partially occluded head

Figure 4.5 shows a case where the two models diverge in the number of heads detected. As can be seen in A.1e, the ground truth of this case contains two annotated heads: a clearly visible head in the upper frame and a second occluded head in the bottom. The baseline model only detected the clearly visible head whilst the circular model detected both heads. Comparing this against the ground truth confirms that the occluded head is a true positive. This is consistent with the extended recall range observed in the PR analysis (section 4.3). The circular model commits to detections at lower confidence threshold, recovering heads that the baseline fails to recover.

which is a big difference
and might explain the
different behaviour for high recall!

On the other hand, in Figure 4.6, a case has been found where both models missed a partially occluded head. Looking at Figure A.1f, two heads were annotated. Both models detect the lower



Circular model (conf. 0.55, 0.26) Baseline model (conf. 0.71)

Figure 4.5: Box model missing a partially occluded head

head, but neither detected the upper head. This shows that both models are not foolproof and can fail to retrieve heads when the visible signal is insufficient to exceed the detection threshold of either model. These cases contribute to the recall ceiling visible in Figure 4.2.

4.4.3 Duplicate detections

Both models occasionally produce two predictions for the same head, which is a known characteristic of $B = 2$ predictors per grid cell in the YOLOv1 architecture. The non-maximum suppresion step described in section 3.8 suppresses most duplicates, but some cases remain visible at the visualisation threshold of 0.2.

For the circular model, duplicate detections show up as multiple circles with different radii on the same head. This is visible on the head in Figure 4.7 where multiple circles are produced despite the head being unoccluded. Inspecting Figure A.1a further, we can also see that both models missed a broccoli detection. The same pattern occurs on the partially occluded head in Figure 4.8, where the circle model produces two circles on the same head unlike the baseline model.

For the baseline model, duplicate detections show up as two slightly offset boxes. This is shown in Figure 4.9, where the heavily occluded head receives two offset boxes whilst the circular model produces a single detection for the same head.

Lastly, Figure 4.10 shows a more complex failure case. The circular model produces three detections: a circle at the top of the frame and two overlapping circles, which are duplicates. The baseline produces one box at the top of the frame and one box in the middle. When looking at the ground truth in FigureA.1d, we see that the detection at the top is a false positive. These figures show us that both models can have false positives and possible duplicate detections.

Circular model (conf. 0.49)

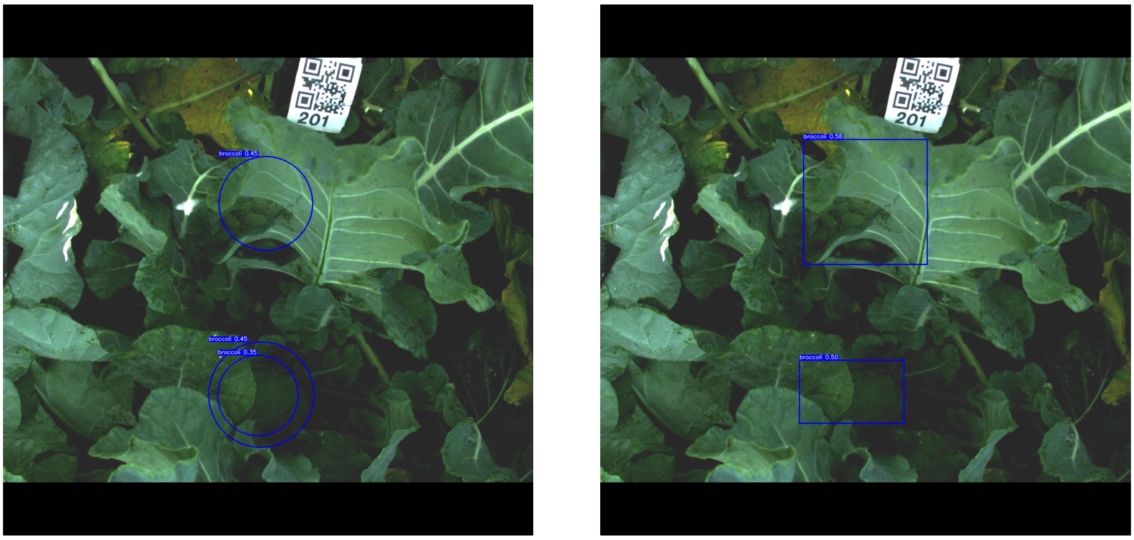
Baseline model (conf. 0.50)

Figure 4.6: Both models missing a partially occluded head

Circular model (conf. 0.29/0.32)

Baseline model (conf. 0.40)

Figure 4.7: Isolated head - duplicate circle detections vs single box and missing detection in both models



Circular model (conf. 0.45, 0.45/0.35) Baseline model (conf. 0.58, 0.50)

Figure 4.8: Partially occluded heads - duplicate circle detections vs single box

Circular model (conf. 0.31) Baseline model (conf. 0.43/0.26)

Figure 4.9: Partially occluded head - single circle detections vs duplicate box

Circular model (conf. 0.29/0.24, 0.30)

Baseline model (conf. 0.21, 0.35)

Figure 4.10: Partially occluded head - duplicate circle detection and wrong detection in both models

Chapter 5

Conclusion

In this thesis we investigated whether the prediction of naturaly circular objects, in particular broccoli's, can be improved by replacing the rectangular bounding box representation in a deep learning detector with a circular one. To answer this question, the YOLOv1 architecture was adapted to predicted circle parameters (cx, cy, r) instead of bounding boxes (x, y, w, h), requiring modifications to the prediction head, target encoding, loss function, non-maximum suppression and evaluation protocol. Both models share an identical ResNet-34 backbone and training configuration, isolating the geometric representation as the sole variable under study.

The circular model achieves a mAP of 0.7598 on the test set at an IoU threshold of 0.5, compared to 0.7038 for the bounding box baseline, an absolute improvement of 0.056 (approximately +8.0%). The precision-recall analysis reveals that this gain is driven primarily by an improvement in recall: the circular model retrieves broccoli heads up to a maximum recall of approximately 0.90, compared to 0.73 for the baseline, while maintaining competitive across the recall range. Qualitative analysis further shows that the circular boundary conforms more naturally to the round contour of broccoli florets, reduces the inclusion of irrelevant background pixels and produces detections on partially occluded heads that the baseline fails to retrieve.

These results provide a positive answer to the central research question: a circle-shaped object detection model does improve the prediction of naturally circular objects compared to a standard rectangular model, both in terms of quantitative detection performance and geometric boundary fit. The findings extend the evidence from Nguyen et al. [7], who demonstrated circular detection on CenterNet-based architecture in the biomedical domain, to the YOLOv1 framwork applied to agricultural crop detection, suggesting that geometric allignment between the model representation and the original object shape is generally applicable principle rather than one specific to a particular architecture or domain.

You need to say something about the much lower confidence (or is that in the discussion?)

Chapter 6

Discussion

6.1 Geometric inefficiency of the rectangular representation

The empirical improvement observed in this thesis is consistent with a geometric argument. For a perfect circular object, with its tightest bounding square, the fraction of the box area that does not belong to the object is exactly $1 - \pi/4 \approx 21.5\%$. This background fraction is an irreducible structural cost of the rectangular representation: it cannot be eliminated by training, data augmentation or by any post-processing step, because it arises from the geometric mismatch between the representation and the object shape itself. The circle representation eliminates this mismatch as a direct consequence of how it is designed. As noted in section 3.9, a perfect circle prediction would achieve a box IoU of only $\pi/4 \approx 0.785$ against its ground-truth bounding square, illustrating that even an ideal circle model would appear to underperform under a rectangular evaluation metric. The consistent use of circle IoU across training, NMS and evaluation in this thesis ensures that the circular model is measured on it own geometric terms, and the +8.0% mAP gain over the baseline reflects a genuine improvement in detection quality.

I don't understand at all
A perfect circle can get a best prediction with the circlemetod with IoU of 1 whereas as you have just remarked the box method gets at most π/4. Therefore the circle method should be performing _better_.

6.2 Structural simplification and efficiency

The circular parameterisation reduces the number of predicted values per predictor from 5 (x_center, y_center, width, height, confidence) to 4 (x_center, y_center, $\sqrt{r}$, confidence), decreasing the prediction tensor from $S \times S \times (5B + C)$ to $S \times S \times (4B + C)$. In the present configuration ($S = 7$, $B = 2$, $C = 1$) this corresponds to a reduction from 539 to 441 output values. This is a negligible amount of parameters relative to the approximately 21.8 million parameters of the ResNet-34 backbone [1]. No inference timing measurements were conducted in this study, so no speed claim is made. However, the structural simplification is a genuine advantage of the circular model: it reduces the degrees of freedom per prediction by one, which may reduce optimisation difficulty during training by presenting a simpler regression target. Whether this translates to a measurable reduction in training time or inference latency would require controlled benchmarking and is thus a direction for future work, particularly in settings with larger grids, more classes, or on resource-constrained hardware.

Number

No it is not negligable: the comparison is between 539 and 441.
They are the ones that end up in the loss function the 22 Million ones are "only latentt"

6.3 Broader application of circular detection

The results of this thesis, together with the findings of Nguyen et al [7] in the biomedical domain and Haas et al. [4] in fluid dynamics, suggest that using circle based models in use cases, where the natural shape of the target object is also round, is a generally applicable design principle. In the biomedical sector, objects such as red blood cells, glomeruli and cell nuclei are inherently circular or almost circular. Nguyen et al. [7] demonstrated the benefit of circular detection on

CenterNet. This thesis shows that this principle holds when applied to a simpler grid-based architecture, broadening the evidence that the advantage is not specific to a particular framework. In fluid dynamics and multiphase flow research, gas bubbles approximate circular or elliptical shapes. The BubCNN framework of Haas et al. [4] adresses this via two-stage pipeline (Faster R-CNN localisation stage combined with a shape regression network). A single-stage circular detector like in this thesis could provide a simpler and more directly trainable alternative for applications where the two-stage overhead is undesirable. In remote sensing and satellite imagery, circular structures such as oil storage tanks, silos and roundabouts appear frequently in aerial and overhead imagery. Yang et al. [?] demonstrated that a circle representation network prediction centre and radius outperforms bounding-box detectors for such targets, explicitly noting reduced background inclusion and rotation invariance as key advantages. Wen et al. [10] similarly proposed a radius detection head for circular targets in remote sensing images, with circle IoU used for evaluation. Considering how similar this methodology is to the approach in this thesis, these independent development further support the principle that geometric alignment between the model predictions and object shape improves the model. In the agricultural domain, the logical extension of this thesis is to apply the circular prediction head to a more recent and higher performing detector. RF-DETR [9], which is the current state-of-the-art object object detection model, is the natural candidate for this. The design of RF-DETR introduces greater architectural complexity than YOLOv1, but the principle of replacing the rectangular output head with a circular one and adapting the loss function accordingly remains similar. Given that YOLOv1's simpler architecture was chosen precisely to isolate the effect of the geometric change, further research with RF-DETR would provide evidence for whether the improvement generalises to a state-of-the-art model under conditions closer to practical deployment.

6.4 Limitations

Several limitations of this thesis should be acknowledged.

First, the confidence thresholds were selected by optimising the F1 score of the baseline model on the validation set and held fixed for the circular model without re-tuning (section 3.8). This choice was made deliberately to ensure that performance differences are attributable to the geometric representation rather than to a combination of threshold optimisation and geometric representation. As a consequence, the reported mAP of the circular model may be a conservative estimate of its maximum achievable performance under its own optimal thresholds.

NO they are not!

Second, the two models are evaluated using different IoU metrics. This ensures each model is judged on its own geometric terms, but it means the mAP values are not directly comparable on a common scale. The +8.0% improvement should be interpreted as evidence that the circular model is better adapted to this task, rather than just a better model.

In the conclusion and discussion it is even more important not to use abbreviations

Third, both models exhibit duplicate detections, arising from $B = 2$ predictors per grid cell in YOLOv1. While NMS suppresses most duplicates, some persist. This is a known structural characteristic of the YOLOv1 architecture and is particularly relevant for the broccoli detection task, because Blok et al. [2] explicitely note that broccoli heads in this dataset grow solitarily and do not overlap each other. This domain knowledge allows one to be able to assume that any two detections with substantial overlap correspond to a duplicate rather than 2 distinct heads. A deduplication post-processing step could exploit this knowledge, by for example surpressing any predicted circle whose centre falls in the radius of a prediction with a higher confidence. This approach could further improve precision and thus possibly be another direction for future work.

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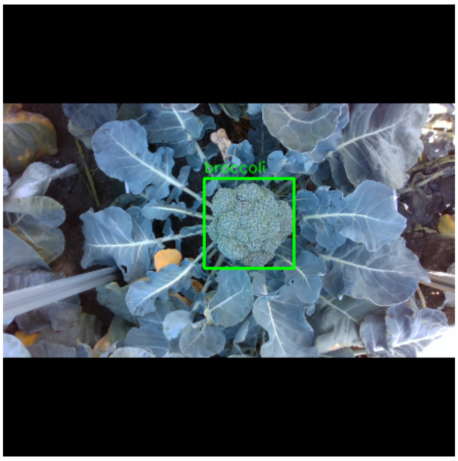
Appendix A

Ground Truth Sample Images

(a) GT of top image of Figure 4.3



(b) GT of bottom image of Figure 4.3

(c) GT of top image of Figure 4.4

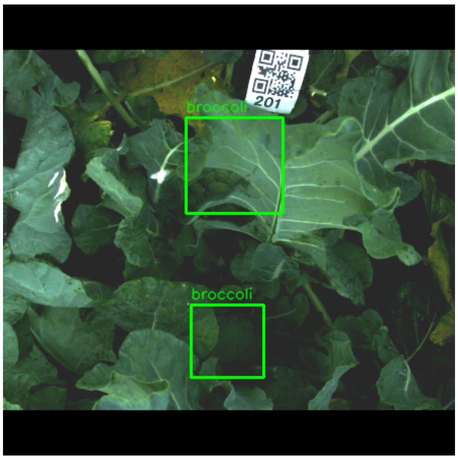
(d) GT of bottom image of Figure 4.4

(e) GT of sample with missing partially occluded head in baseline model (4.5)

(f) GT of sample with missing partially occluded head in both models (4.6)

Figure A.1: Ground truth annotations for all qualitative result figures (cont.)

(a) GT of Figure 4.7



(b) GT of Figure 4.8

(c) GT of Figure 4.9, looking at duplicates

(d) GT of Figure 4.10 with duplicates and wrong detections

Figure A.1: Ground truth annotations for all qualitative result figures.